

INVITED TALK / ABSTRACT

A Nonlinear Operator Approach to Black Hole Solution Classes in the Ernst Equation

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Abstract

The Ernst equation arises from the dimensional reduction of the vacuum Einstein equations in the presence of two spacelike Killing vector fields and plays an important role in the description of stationary axisymmetric spacetimes. In this work, we introduce a profile–phase decomposition of the Ernst potential that naturally induces a system of nonlinear differential operators encoding its intrinsic analytic structure. Rather than relying on a prescribed solution-generating ansatz, the induced operator system determines a broad class of admissible solutions through nonlinear differential constraints, providing an intrinsic description of the corresponding solution space. Moreover, we show that suitable rigidity and asymptotic conditions select the classical black hole families within this larger solution space. In particular, the full two-dimensional phase geometry contains the Kerr–NUT sector, while one-dimensional totally geodesic reductions recover the static sector, including Schwarzschild and its Zipoy–Voorhees deformation. This work is in collaboration with Felipe Poblete.

Keywords: nonlinear operator approach; black hole solution classes; Ernst equation; vacuum Einstein equations; nonlinear differential operators; rigidity; asymptotic conditions; Kerr–NUT sector